



WEEKLY PROBLEM

Energy Optimization in an Autonomous Electric Delivery Network

EXERCISE.

A logistics company operating in a smart city deploys a fleet of autonomous electric delivery vehicles. The instantaneous energy consumption rate (in kWh) of a representative vehicle during its route is modeled by:

$$E'(t) = (2 + 0.5t)e^{-0.1t} + 3\sin\left(\frac{\pi t}{6}\right)$$

where: t is time in hours ($0 \leq t \leq 12$), the first term represents **battery discharge under load and efficiency decay**, the second term captures **periodic acceleration cycles due to traffic signals**.

Additionally, due to regenerative braking and adaptive AI control, the efficiency, the **effective energy stored back into the battery** over time is modeled by:

$$R(t) = \int_0^t (1.2e^{-0.05\tau} \cos(\frac{\pi\tau}{6})) d\tau$$

1. Net Energy Consumption Over Route

Determine the **total net energy consumed** by the vehicle over the 12-hour operational window, taking into account both consumption and recovery.

2. Critical Load Analysis

At what time(s) does the **instantaneous energy rate reach a local maximum**?

3. Sustainability Constant Violation

- Determine the time interval where:

$$E'(t) - R'(t) > 4$$

- Based on your findings, propose whether route rescheduling, adaptive speed control or battery upgrade is the most mathematically justified intervention.



(Scan this to submit solution!)

Top Submissions for last Week's Problem (WP2604):

- Kumar Raushan Mehta** - AD25B1015
- Adelly Sai Teja** - CS25B1003
- A Madhava Selvan** - MC25B1001
- Giripogu Vikas Joel** - AD25B1010